

Model Paper - Database Management Systems

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Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

In a university database system, there are four main entities: 'Professor', 'Course', 'Student', and 'Department'. The 'Professor' entity has attributes: ProfessorID, Name, and Email. The 'Course' entity includes CourseID, CourseTitle, and Credits. The 'Student' entity consists of StudentID, Name, and EnrollmentDate, while the 'Department' entity is characterized by DepartmentID, DepartmentName, and OfficeLocation. The 'Department' offers multiple 'Courses', and each 'Course' can have multiple 'Students' enrolled. Furthermore, there is an ISA hierarchy where 'Professor' can be a 'VisitingProfessor' or a 'FullTimeProfessor', with specific attributes for each such as VisitingProfessorID and ResearchArea for the 'VisitingProfessor' and OfficeHours for the 'FullTimeProfessor'

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R : $F = \{AB \rightarrow C, B \rightarrow D, A \rightarrow E, CD \rightarrow B, E \rightarrow A\}$. Analyze the relation structure, identifying keys and normalization compliance.

- a) Find all the keys in relation R using the attribute closure method. (4 marks)
- b) Is R in 3NF? Give reasons for your conclusion. (3 marks)
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is in the process of developing a comprehensive database system to manage its student and course information efficiently. Various roles are given to different team members for the administration of this database system. Alex is the chief database administrator overseeing the entire system's creation and functionality. Jamie and Chris are junior database administrators working under Alex's supervision. Jamie manages the student records database, while Chris looks after the course catalog database. Taylor is a database developer reporting to Jamie and Chris. Taylor's responsibilities include designing and implementing database schemas for students and courses. Morgan and Reese are data entry operators responsible for entering student details and course data into the system.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

- b) Code Segment:

```
String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql);
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

- c) Briefly explain the options available for transferring data between two tables within the same database or across different databases. (2 marks)

- d) Briefly explain how authentication and authorization are achieved in MySQL. (3 marks)

e) A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects.

Michael is a data entry operator responsible for entering and updating student enrollment data.

- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role. (5 marks)
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a Library: Book (bookId: int, title: varchar(100), author: varchar(50), publicationYear: int) Member (memberId: int, name: varchar(100), membershipDate: date, email: varchar(50)) Loan (loanId: int, memberId: int, bookId: int, loanDate: date, returnDate: date, status: varchar(20)) Fine (fineId: int, loanId: int, amount: decimal(10,2), paymentStatus: varchar(20)) The 'Book' table stores information about books available in the library, including unique book ID, title, author, and publication year. The 'Member' table retains details of library members, incorporating member ID, name, membership date, and email. The 'Loan' table records book loans to members and includes loan date, return date, and the status of the loan. The 'Fine' table documents fines associated with loans, detailing amount and payment status. The 'Member' table holds details about library members, such as their unique member ID, first name, last name, email, phone number, and address. The 'Loan' table manages book loans, with each loan having a unique ID and being associated with a specific book and member. It records the loan date, due date, and return date. The 'Fine' table contains information about fines incurred by members for late returns or other penalties. It includes a fine ID, member ID, fine amount, and payment status.

a) Write SQL Queries to perform the following:

- i. Find the name and email of members who have a loan that is currently overdue. (4 marks)
 - ii. Find the member who has the highest total amount compared to all other members. Find the memberId and name. (6 marks)
 - iii. Find the title and name from Book, Member, and Loan where Loan.loanId is not null. (7 marks)
- b) Create a function to calculate the total fine amount for a given member. This function should account for member with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)
- c) Create a trigger that automatically updates the 'amount' column in the 'Fine' table whenever a new fine record is added, or an existing fine record is updated. The 'amount' column should contain the total fine amount for each member. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by member. The member pays only 50% of the total fine amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)